

MLOps Foundations Executive Brief*SRAI Book 5 Lesson 1 | PU-B05-C01 | Executive Brief v0.1**Mbaye Kebe***Decision**

Treat MLOps as the operating and governance system for machine learning, not as deployment automation. Require an evidence chain from accepted data through retirement. Keep consequential release blocked until a named authority approves the declared use.

Why this matters

A model can perform well in development and still fail through changed data, inconsistent transformations, fragile infrastructure, silent degradation or unauthorized use. Public institutions must preserve records that remain understandable after staff, vendors and platforms change.

Minimum operating controls

Control	Executive question
Purpose	What decision may this system support, and what uses are prohibited?
Evidence	Can an independent reviewer reconstruct the candidate and evaluation?
Accountability	Who owns data, evaluation, operation, risk and approval?
Release	Which mandatory conditions block promotion?
Monitoring	Who receives alerts and what action follows?
Recovery	Can the complete known-good service be restored?
Retirement	How will traffic, credentials and downstream dependencies be closed?

Cross-sector implications

Sector	Primary concern	Required safeguard
Agriculture	Seasonal and spatial change	Time-aware validation and field verification
Education	Support indicators becoming punitive	Human review and prohibited-use controls
Health	False reassurance or unsafe delay	Clinical validation, escalation and conservative failure
Habitat	Community exclusion or spatial overclaim	Valid geographic resolution and contestability

Management actions

- Approve an explicit intended-use statement and prohibited uses.
- Fund reproducibility, monitoring and rollback as recurring operations.
- Separate model building from independent evaluation and production approval.
- Require immutable release manifests and authenticated decisions.
- Review systems when data, purpose, population, policy or infrastructure changes.

Lesson evidence

The companion notebook builds a synthetic agricultural-risk classifier and records environment, configuration, accepted data, deterministic splits, baseline and candidate metrics, regional slices, artifact hashes, contract tests, drift evidence, a model card, release manifest and rollback record. The canonical run keeps governance approval false, so release remains correctly blocked.

Executive conclusion

MLOps creates value when it makes system behavior reconstructable, observable and reversible. Governance creates the authority and limits for use. Institutions need both. A successful pipeline provides evidence for a decision; it does not make the decision.